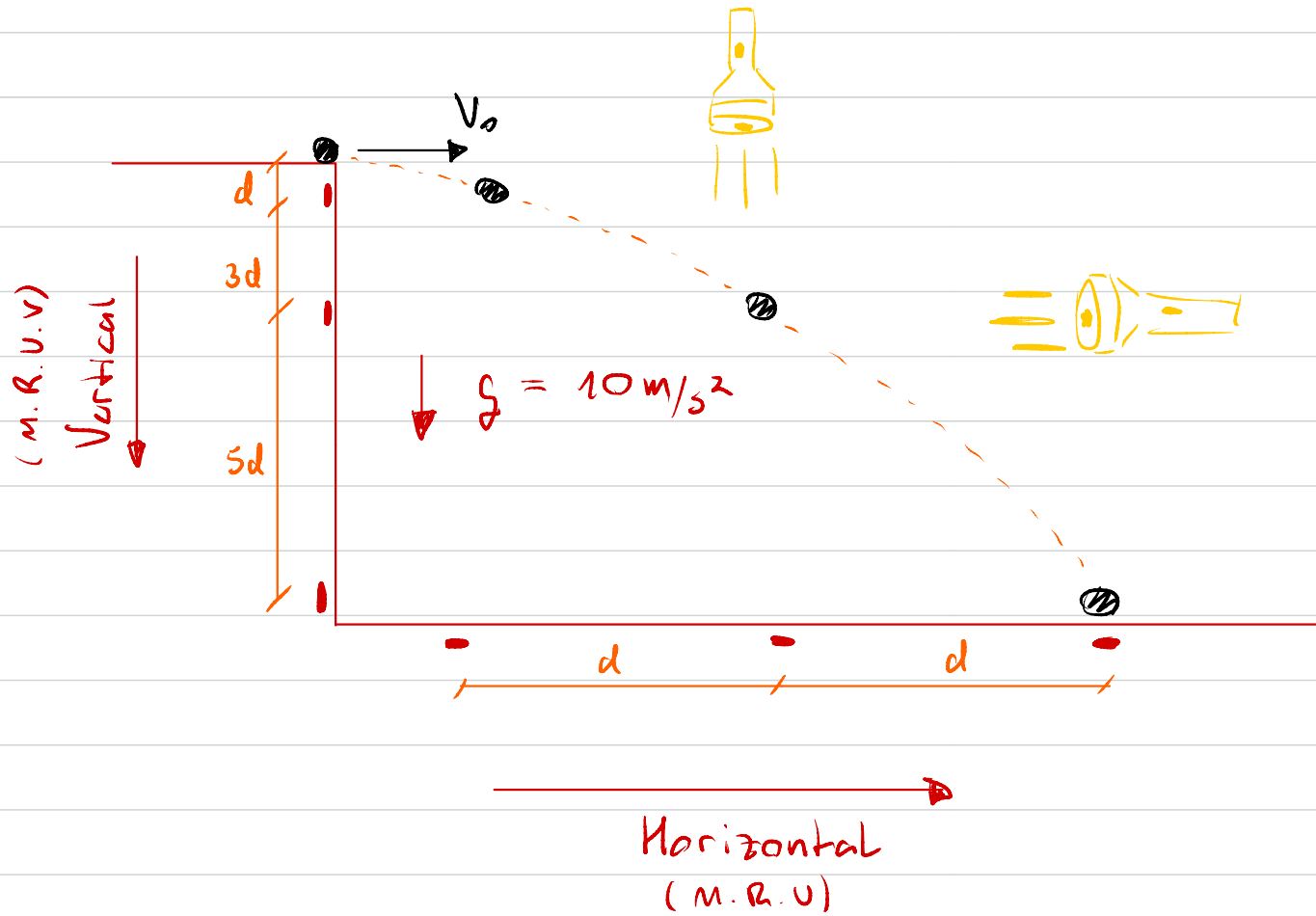
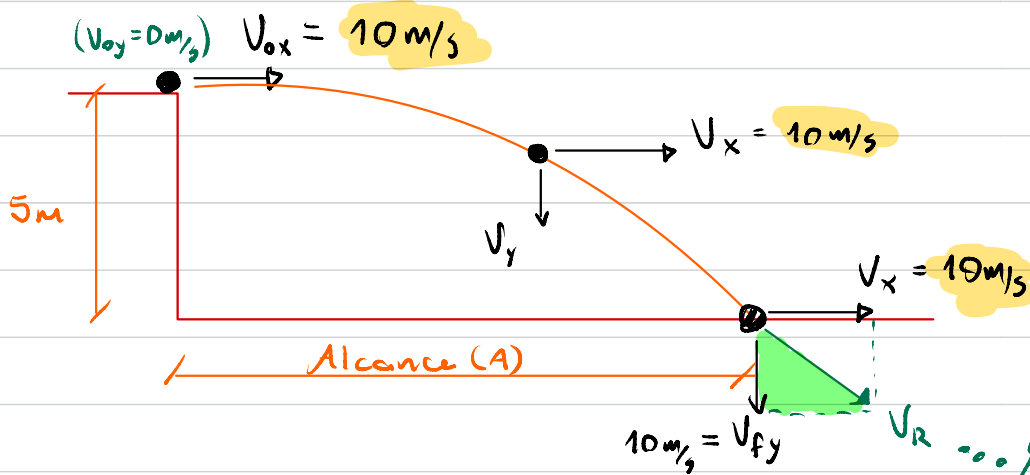


Lançamentos

① Lançamento Horizontal



Exemplo



1) Tempo de queda

Vertical:



$$\Delta s = v_i t + \frac{at^2}{2} \rightarrow H = 0 \cdot t + 8 \cdot \frac{t^2}{2}$$

$$5 \text{ m} = 0 \cdot t + 10 \cdot \frac{t^2}{2}$$

$$\frac{5}{5} = t^2 \therefore \boxed{t = 1 \text{ s}}$$

$$\boxed{t = \sqrt{\frac{2H}{g}}}$$

$$V_R^2 = V_x^2 + V_y^2$$

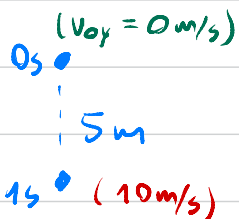
$$V_R^2 = 10^2 + 10^2$$

$$\boxed{V_R^2 = \sqrt{2 \cdot 100}}$$

$$\boxed{V_R = 100 \cdot \sqrt{2} \text{ m/s}}$$

2) V_{fy} após a queda?

Vertical



$$\begin{aligned} & \bullet V_{0y} = 0 \text{ m/s} \\ & \bullet g = 10 \text{ m/s}^2 \\ & \bullet t = 1 \text{ s} \end{aligned}$$

$$V_{fy} = V_{0y} + a \cdot t$$

$$V_{fy} = 0 + 10 \cdot 1$$

$$\boxed{V_{fy} = 10 \text{ m/s}}$$

3) Alcance

Horizontal → m.R.U

$$S = v \cdot t$$

↓ ↓

$$A = v_x \cdot t$$

$t = t_{\text{queda}}!$

$$A = 10 \cdot 1s$$

$$A = 10m$$

$$A = v_x \cdot \sqrt{\frac{2 \cdot H}{g}}$$

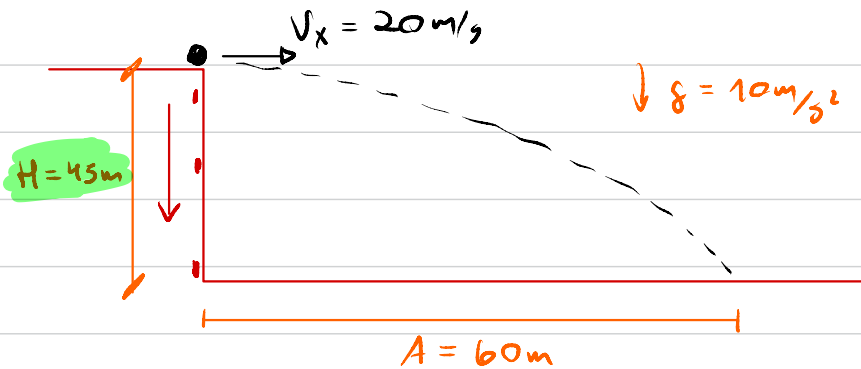
Altura

Gravidade

Velocidade
Horizontal
de lançamento

* Note que a massa não influencia no alcance!

Ex: Determine a altura do Lançamento Horizontal à seguir:



Horizontal

$$A = v_x \cdot t$$

$$60 = 20 \cdot t$$

$$t = 3s$$

t_{queda}

Vertical

